Introduction to Software Engineering

Project Idea & Team Contract

The student team is required to complete the **Project Idea & Team Contract** documentation for the assigned course project, following the attached template.

Software Engineering Department
Faculty of Information and Technology
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Project Idea & Team Contract

Objectives

This document presents the miniTA project idea for approval and records the working agreement of the four-member team.

- ✓ Describe the project at a high level before detailed requirements are written in PA2.
- ✓ Define the scope that the team can complete during the course implementation period.
- ✓ Agree on communication, responsibilities, work standards, and AI usage.

1 Team Information

Item	Value
Team ID	T01
Team name	miniTA
Team lead	Nguyễn Cao Triết
Communication channel	Discord

Student ID	Full name	Email	Main role	Backup role
24127572	Nguyễn Cao Triết	nctriet2441@clc.fitus.edu.vn	*<Team lead / PM>*	*<Developer>*
24127447	Phùng Gia Mẫn	pgman2413@clc.fitus.edu.vn	*<Business analyst>*	*<Tester>*
24127323	Tống Vân Anh	tvanh2418@clc.fitus.edu.vn	*<Designer>*	*<Developer>*
24127207	Nguyễn Thục Nghi	ntnghi2426@clc.fitus.edu.vn	*<Developer>*	*<Tester>*

2 Project Idea Overview

2.1 Project Name and One-Sentence Pitch

- miniTA is a web-based applicant tracking system that helps small HR teams collect applications, compare text-based PDF resumes against transparent job-specific criteria, and keep candidates informed of their application status.

2.2 Problem and Motivation

- Small companies often manage recruitment through fragmented tools, resulting in scattered data and overlooked candidates. Manually reviewing unstructured PDF resumes without predefined rubrics leads to inconsistent evaluations. Meanwhile, candidates lack visibility into their application status, forcing recruiters to waste time answering status inquiries instead of evaluating skills.
- To solve this, miniTA centralizes the workflow around each job posting. Recruiters can define standard rubrics and share a single application link. The system automatically extracts text from submitted PDFs and calculates an explainable match score, clearly showing matched skills, missing requirements, and supporting evidence. Simultaneously, candidates can securely track their application status via an email magic link.
- Importantly, miniTA is designed strictly as an internal tracking workspace, not a public job board or social network. It acts solely as a review aid; the system never automatically rejects, shortlists, or hires candidates. All final decisions remain in the hands of human reviewers, supported by transparent, skills-based evidence.

2.3 Target Users

User group	What they need from the system
Recruiters/HR	Create job postings, configure evaluation criteria, receive applications in one workspace, review resumes and matching evidence.
Candidates	Apply through a simple public link, upload a PDF resume, check their submitted information and application status.

2.4 Main Features

#	Feature	Short description	Must-have?
1	Account & Role Management	Recruiters log in and access specific records. Hiring managers get limited review access.	Yes
2	Job Posting Management	Create, edit, publish, close, and share job postings with public links.	Yes
3	Rubric builder & AI job-intake assistant	Recruiters define required/preferred skills, experience, weights, and screening questions. AI may draft these from a pasted JD, but the recruiter must approve them.	Yes
4	Candidate application portal	Candidates apply through the public link, provide contact information, answer screening questions, and upload one PDF CV/resume.	Yes
5	Candidate status portal	Candidates receive a time-limited email magic link to securely view their submitted application and current status.	Yes
6	Automated CV extraction & transparent matching	System extracts text from PDF CVs, calculates a match score (%) against the JD/approved rubric, and shows a breakdown of matched/missing skills with text evidence.	Yes

7	Applicant inbox & human review workflow	Recruiters search, filter, sort, add notes, submit human scorecards, and update candidate statuses.	Yes
8	Recruitment dashboard	Recruiters view application counts by job and status, score distribution, and basic pipeline conversion statistics.	Yes
9	Inbound application email	Candidates send a resume to a job-specific email address, and the system automatically creates an application from the attachment.	No
10	Interview scheduling & email templates	Integrated email notifications allowing recruiters to send reusable status emails or interview invitations directly from the system.	No

2.5 Similar Products

- ***SmartRecruiters:*** An enterprise ATS with complex CRM, onboarding, and job distribution features. miniTA is much simpler, focusing strictly on the core tracking workflow: public links, screening, transparent matching, and status tracking.
- ***Workable/Lever:*** Full recruitment platforms with sourcing and calendar integrations. miniTA excludes these advanced features to fit the course project scope.
- ***TopCV/VietnamWorks:*** Public job boards and candidate marketplaces. miniTA is not a marketplace; it is an internal workspace for companies to process direct applications via their own shared links.

2.6 Platform and Technology Ideas

- ***Platform:*** Responsive Web Application.
- ***Frontend:*** ReactJS and TypeScript.
- ***Backend:*** Python and FastAPI.
- ***Database:*** PostgreSQL.

- ***Data Processing:*** pdfplumber or PyMuPDF for PDF text extraction; pandas for string/data manipulation. Rule engine for matching.
- ***AI assistance:*** LLM API with validated structured JSON output

2.7 Feasibility, Scope and Open Questions

- ***Feasibility:*** The 4-member team can implement the must-have scope within 3 weeks. PDF extraction and matching APIs are highly feasible using Python.
- The implementation can be divided into three stages:
  - Stage 1 (Foundation): Database schema, role-based authentication, Job CRUD, and private file uploads.
  - Stage 2 (Core Review): PDF text extraction, rule-based matching, applicant inbox, and status pipeline.
  - Stage 3 (Completion): Candidate magic-link portal, dashboard, AI rubric suggestions, and testing.
- ***Out of scope:*** OCR for scanned resumes, social networking, job-board distribution, calendar/video integrations, and automatic hiring/rejection decisions are excluded from the first release.
- ***Risks and mitigations:***
  - PDF Extraction Failures: Complex layouts may not extract well; the system will only support selectable-text PDFs initially and flag failures for manual review.
  - Data Privacy: Resumes are sensitive; mitigated by requiring candidate consent, using fictional/consented test data, and enforcing strict authorization.
  - Biased Matching Scores: Addressed by using transparent rubrics, showing exact text evidence, and keeping humans in control of all decisions.
  - LLM Inconsistencies: AI is restricted to drafting rubrics and summaries only, with validated outputs and manual fallback options.

- Scope Creep: Prevented by treating all non-essential features as stretch goals and prioritizing the core end-to-end workflow.

- ***Questions for TA:***

  1. If the 'CV Matching' algorithm or 'PDF Extraction' feature occasionally misreads the format or misses some skills, will we lose a lot of points, or is it enough that the basic logic works correctly?

  2. Will you evaluate individual contribution scores directly based on the number of lines of code or Pull Requests on GitHub?

  3. During the final demo, will you use your own sample CVs to test our system, or should our team prepare our own CVs for the demo?

3 Team Contract

3.1 Goals and Expectations

- The team aims to achieve a final project mark of **at least 8.5/10**. Each member commits to an average of six hours per week, completing assigned work, communicating blockers early, and reviewing teammates' work.

3.2 Communication

- Main channel: Discord link: **https://discord.gg/eb4rXqDG**
- Expected response time: Within 24 hours on weekdays.
- Project records: GitHub for code; Jira for tasks; Google Drive for documents.

3.3 Meetings

- Weekly status meeting with the TA: at 9pm Monday.
- Internal team meeting: at 8pm every Thursday.
- Attendance rule: a member informs the team before a missed meeting. Meeting notes record absences, decisions, action items, and changed deadlines.
- Weekly status report deadline: at 5pm Monday (it can be on the same day as weekly status meeting with the TA).

3.4 Roles and Responsibilities

- PM and BA (Triết & Mẫn): Manage scope, requirements, tasks, and PA submissions.
- Designer (Anh): UI/UX design, validation.
- Developer (Nghi with Triết & Vân Anh): Frontend and Backend implementation, integration, database.
- Testing: Mẫn & Nghi but other members can also assist with this roles and QA as backup.

3.5 Work Standards

- All tasks must have an assigned owner, a due date, and clear acceptance criteria, with the goal of completing assigned work at least two days before deadlines.
- Code changes are managed strictly through feature branches and pull requests.
- A feature is only considered complete once it meets its acceptance criteria, passes testing, includes updated documentation, and receives peer review.
- Furthermore, to protect privacy, the team will solely use fictional or consented test data, ensuring no real candidate information is exposed.

3.6 Use of AI Tools

- AI tools (ChatGPT, Copilot,...) may be used for brainstorming, drafting, and small code suggestions. AI output must be reviewed, tested, and verified. Significant use must be recorded in the AI log.

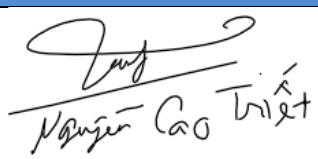
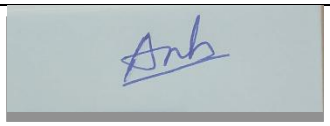
3.7 Decision Making

- The team first seeks consensus for scope, architecture, and design decisions. If consensus is not reached after discussion, a simple majority vote decides the issue. In a tie, the Project Manager makes the final decision after documenting the alternatives, project constraints, and reason for the decision. Any decision that changes a PA requirement or deadline is recorded in meeting notes and on the task board.

3.8 Handling Conflicts and Missed Work

- Step 1: Discuss directly within 48 hours to agree on a fix.
- Step 2: Discuss in a team meeting and record in the weekly report.
- Step 3: Inform the TA if the issue persists.

3.9 Signatures

Student ID	Full name	Signature	Date
24127572	Nguyễn Cao Triết		13/10/2026
24127447	Phùng Gia Mẫn		13/10/2026
24127323	Tống Vân Anh		13/10/2026
24127207	Nguyễn Thục Nghi		13/10/2026

4 AI Usage Log

#	Member	Tool	Task / purpose	How the output was verified and changed
1	All team members	Codex	Researched publicly described SmartRecruiters features to identify relevant applicant-tracking patterns for miniTA, including public application links, candidate status tracking, screening questions, and recruiter workflows.	The team checked the official SmartRecruiters pages cited in this report, selected only features feasible for miniTA, and excluded enterprise modules such as CRM, job-board distribution, scheduling, offers, and onboarding.
2	All team members	Gemini	Drafted the initial summaries for the "Problem and Motivation" and "Similar Products" sections to ensure concise and professional wording.	Reviewed by the team lead, manually adjusted to match the specific scope of miniTA (e.g., emphasizing the email magic link), and translated/refined to align with the final report structure.